## Table of expected frequencies:

| Number of heads | 0 | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|---|
| Expected frequencies | 20 | 100 | 200 | 200 | 100 | 20 | 640 |

**Step 3 : Level of significance**

$\alpha = 5\%$

**Step 4 : Test statistic**

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

**Step 5 : Calculation**

The test statistic is computed as under:

| Observed frequency $(O_i)$ | Expected frequency $(E_i)$ | $O_i - E_i$ | $(O_i - E_i)^2$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|---|---|---|---|---|
| 19 | 20 | –1 | 1 | 0.050 |
| 99 | 100 | –1 | 1 | 0.010 |
| 197 | 200 | –3 | 9 | 0.045 |
| 198 | 200 | –2 | 4 | 0.020 |
| 105 | 100 | 5 | 25 | 0.250 |
| 22 | 20 | 2 | 4 | 0.200 |
| | | | Total | 0.575 |

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

$$= 0.575$$

**Step 6 : Critical value**

Degrees of freedom = $k - 1 - s = 6 - 1 - 1 = 4$

Critical value for *d.f* 4 at 5% level of significance is 9.488 *i.e.*, $\chi^2_{4,0.05} = 9.488$

**Step 7 : Decision**

As the calculated $\chi_0^2 (=0.575)$ is less than the critical value $\chi^2_{4,0.05} = 9.488$, we do not reject the null hypothesis. Hence, the fitting of binomial distribution is appropriate.



12th_Statistics_EM_Unit_2.indd 64 1/28/2020 7:10:08 PM